Week 5 Lab 2: Research Activity Log Use-Case Descriptions

Purpose

These use cases define the observable behaviour required for the Research Activity Log introduced in Week 5 Lab 2.

The use cases describe user goals and system responses. They do not prescribe MongoDB, a document structure, repository classes, or any other persistence implementation.

Scope

Core Use Case

- `W5-UC03`: View Student Research Activities

Extension Use Cases

- `W5-UC04`: Record Research Activity
- `W5-UC05`: View Research Activity Details

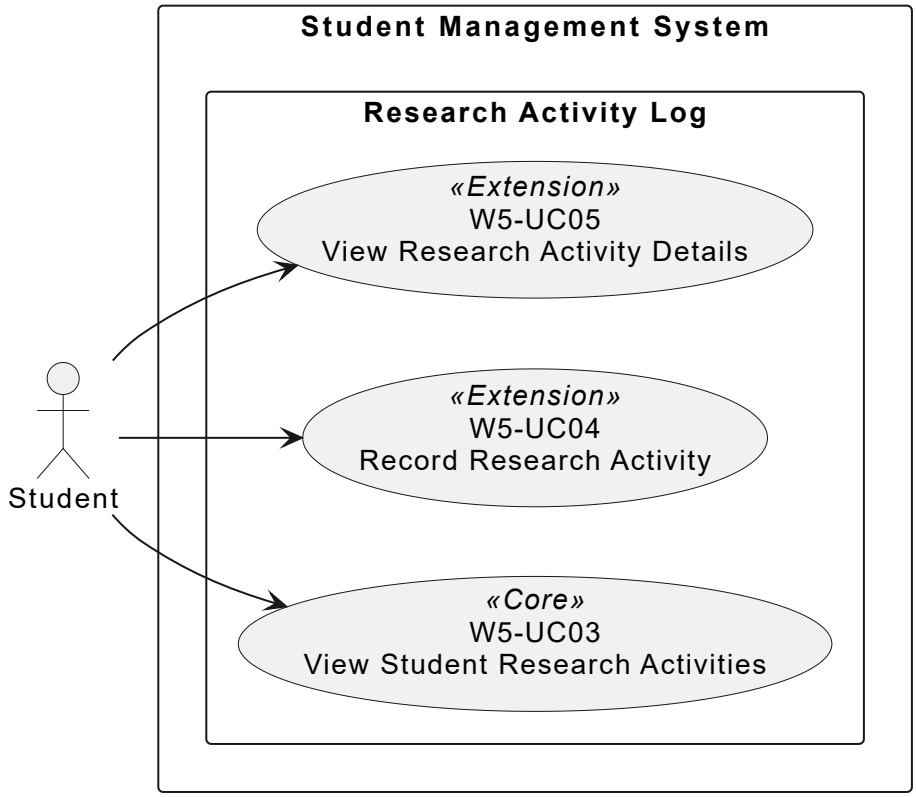
The core lab requires only `W5-UC03`. Implement the extension use cases only after the core document model, repository, service, controller, view, and persistence checks work.

Use-Case Summary

Use Case ID	Use Case	Primary Actor	Classification
W5-UC03	View Student Research Activities	Student	Core
W5-UC04	Record Research Activity	Student	Extension
W5-UC05	View Research Activity Details	Student	Extension

Use-Case Diagram

Week 5 Research Activity Log Use Cases



Core Use-Case Description

W5-UC03: View Student Research Activities

Primary Actor: Student
Classification: Core

Goal

View the research activities recorded for the selected student in chronological order.

Preconditions

- The Student Management System is available.
- The Student can access the Research Activity Log.
- A student number has been supplied by the application or selected for the exercise.

Trigger

The Student selects **Research Activities**.

Main Success Scenario

1. The system receives the student number.
2. The system finds one or more Research Activities associated with that student number.
3. The system orders the activities by activity date, from earliest to latest.
4. The system displays the student number.

5. For each activity, the system displays:
    - Activity type
    - Title
    - Activity date
    - Tags, where recorded
    - Selected detail information appropriate to the activity type
6. The Student reviews the Research Activities.

Alternative Flow A1: No Research Activities Found

At the step where the system searches for Research Activities:

1. No Research Activities are associated with the supplied student number.
2. The system displays **No research activities found**.
3. No Student or Research Activity information is changed.
4. The use case ends.

Postconditions

Success

- The Research Activities associated with the student have been displayed in chronological order.
- No Student or Research Activity information has been changed.

No Research Activities Found

- No Research Activity information has been displayed.
- No Student or Research Activity information has been changed.
- A **No research activities found** message has been displayed.

Business Rules

- A Research Activity is associated with exactly one student number.
- A student may have zero, one, or several Research Activities.
- Activities are ordered from earliest to latest activity date.
- Every activity has an activity type, title, activity date, and creation date.
- Tags are optional.
- Activity-specific details may differ according to activity type.
- Viewing Research Activities must not modify stored data.

Design Decisions

Decide:

- Which Student page provides the **Research Activities** link.
- How activity dates are formatted.
- How tags are displayed.
- Which detail fields appear in the list for each activity type.
- Whether the list shows a short summary or all activity-specific details.
- Whether activities with the same date require a secondary ordering rule.

- Whether an unknown student number is distinguished from a known student with no activities.

For the core implementation, an empty activity query may produce **No research activities found** for both cases. If you introduce a separate **Student Not Found** outcome, document it as an additional alternative flow.

Data Required

The use case requires:

- A student number.
- Research Activities associated with that student number.
- A query that orders matching activities by activity date.
- A representation that supports shared and activity-specific fields.

The persistence technology and document structure are not part of the observable use-case behaviour.

Extension Use-Case Descriptions

W5-UC04: Record Research Activity

Primary Actor: Student
Classification: Extension

Goal

Record a new Research Activity in the student's Research Activity Log.

Preconditions

- The Student Management System is available.
- The Student can access the Research Activity Log.
- A student number has been supplied by the application or selected for the exercise.

Trigger

The Student selects **Record Research Activity**.

Main Success Scenario

1. The system displays the Research Activity creation form.
2. The Student selects an activity type.
3. The Student enters the common activity information:
    - Title
    - Activity date
    - Tags, where applicable
4. The system displays or accepts the detail fields appropriate to the selected activity type.
5. The Student enters the activity-specific details.
6. The Student submits the activity.
7. The system validates the supplied information.
8. The system creates the Research Activity with:

    - A unique activity identifier
    - The selected student number
    - The selected activity type
    - The supplied common and activity-specific information
    - A creation date and time
9. The system stores the Research Activity.
10. The system displays the created activity or the updated Research Activity list.

Alternative Flow A1: Required Information Missing

At the validation step:

1. One or more required common or activity-specific values are missing.
2. The system does not create the Research Activity.
3. The system redisplays the form with a validation message.
4. Previously entered usable values remain available for correction.
5. The Student may correct the information and submit again, or cancel the operation.

Alternative Flow A2: Unsupported Activity Type

At the step where the selected activity type is processed:

1. The activity type is not supported by the current implementation.
2. The system does not create the Research Activity.
3. The system displays **The selected research activity type is not supported**.
4. The Student may select a supported type or cancel the operation.

Postconditions

Success

- One new Research Activity has been stored for the selected student number.
- The activity is available in the student's Research Activity list.

Failure

- No Research Activity has been created.
- Existing Research Activities remain unchanged.

Business Rules

- Every Research Activity has a unique identifier.
- Every Research Activity is associated with exactly one student number.
- Activity type, title, activity date, and creation date are required.
- Activity-specific required fields depend on the selected activity type.
- The creation date is assigned by the system.
- Tags may be omitted.
- Recording one activity must not modify another activity.

Design Decisions

Decide:

- Which activity types are supported by the implementation.
- Which fields are required for each supported type.
- Whether users can enter new tags or select existing tags.
- Whether the student number is selected, supplied by the route, or obtained from the current user.
- Whether the success response shows details or returns to the list.
- How creation time is represented and displayed.

Data Required

The use case requires:

- Common activity information.
- Activity-specific details.
- A student reference.
- A unique activity identifier.
- A creation timestamp.

A document model may store shared and activity-specific values together, but that implementation choice is not part of the use-case scenario.

W5-UC05: View Research Activity Details

Primary Actor: Student
Classification: Extension

Goal

View all available information for one Research Activity.

Preconditions

- The Student Management System is available.
- The Student can access the Research Activity Log.
- A Research Activity identifier has been selected or supplied.

Trigger

The Student selects **Details** for a Research Activity.

Main Success Scenario

1. The system receives the Research Activity identifier.
2. The system finds the corresponding Research Activity.
3. The system displays:
    - Activity type
    - Title
    - Activity date
    - Creation date and time
    - Tags, where recorded

    - All stored activity-specific details
4. The Student reviews the Research Activity details.

Alternative Flow A1: Research Activity Not Found

At the step where the activity is retrieved:

1. No Research Activity matches the supplied identifier.
2. The system displays **Research activity not found**.
3. No information is changed.
4. The use case ends.

Postconditions

Success

- The selected Research Activity has been displayed.
- No information has been changed.

Research Activity Not Found

- No Research Activity details have been displayed.
- No information has been changed.
- A **Research activity not found** message has been displayed.

Business Rules

- Activity details are selected using the unique activity identifier.
- The displayed detail fields depend on the stored activity type and document content.
- Missing optional details are omitted or displayed consistently.
- Viewing activity details must not modify stored data.

Design Decisions

Decide:

- How lists and nested detail values are displayed.
- How missing optional values are handled.
- Whether technical document identifiers are shown to the user.
- Whether the page provides navigation back to the student's activity list.

Data Required

The use case requires:

- A Research Activity identifier.
- The complete stored Research Activity document.

Seed-Data Expectations

The Lab 2 seed data should allow students to test the core use case without manually creating records first.

Include at least:

- Two different activity types for `S100001`.
- One Research Activity for `S100002`.
- One known student with no Research Activities.

The seed data should demonstrate:

- Shared fields used by every activity.
- Different activity-specific detail structures.
- Tags represented as a collection.
- Chronological ordering.
- The **No Research Activities Found** alternative flow.

Use fictional, non-sensitive test data.

Existing Module Behaviour

The Research Activity Log is an additional business module.

Implementing it must not change the observable behaviour of:

- Student and Enrolment Management using relational persistence.
- Academic Supervisor Management.
- Existing public module contracts.

If Research Activity Log confirms that a student exists, it should use a deliberately exposed Students-module contract rather than accessing the Students-module DAO or `DbContext` directly.